Project Proposal & System Methodology: Hybrid RAG Customer Support

Executive Summary

This document defines the project scope, technical specifications, design methodology, and evaluation framework for the **Hybrid RAG & Fine-Tuned Customer Support System**. The objective is to build an intelligent, policy-grounded customer service assistant capable of:

1. Routing user queries to precise domain intents via a fine-tuned Intent Router LLM.

2. Retrieving relevant Standard Operating Procedures (SOPs) from a vectorized knowledge base (ChromaDB + SentenceTransformers).

3. Synthesizing accurate, hallucination-free customer support responses grounded strictly in retrieved corporate policies.

Task 1.1: Define Project Scope

1.1.1 Target Use Case & System Boundaries

- **Domain**: E-Commerce & SaaS Customer Service Automation.
- **Input Specification**: Unstructured natural language query from customers (e.g., *"How do I track my order if it's delayed?"* or *"Can I get a refund after 30 days?"*).
- **Target Routing Schema (JSON Output)**:

`json

{

"intent": "shipping_delays",

"category": "LOGISTICS"

}

`

- **Knowledge Base Scope**: 13 corporate SOP Markdown documents:
- `refund_policy.md`, `shipping_delays.md`, `password_reset.md`, `technical_troubleshooting.md`, `escalation_matrix.md`, `account_recovery.md`, `billing_disputes.md`, `data_privacy.md`, `order_tracking.md`, `payment_methods.md`, `product_return.md`, `subscription_cancellation.md`, `working_hours.md`.
- **Boundaries & Exclusions**:
- The system will NOT perform transaction execution (e.g. directly debiting credit cards or changing database states).
- Out-of-scope queries (unrelated to the 13 corporate SOPs) will trigger an explicit fallback policy.

1.1.2 Success Criteria & Target Metrics

Metric	Definition	Baseline Target	Solution V1 (RAG)	Solution V2 (Hybrid RAG)
:---	:---	:---:	:---:	:---:
Format Adherence Rate	% of generated outputs parsing as valid JSON	>= 60%	>= 80%	**>= 98%**
Intent Routing Accuracy	Exact match % between predicted vs ground truth intent	>= 50%	>= 75%	**>= 95%**

ROUGE-1 / ROUGE-L	N-gram overlap against SOP ground truth reference	Baseline	+25% relative	**+50% relative**
BLEU Score	Precision-based n-gram overlap vs SOP ground truth	Baseline	+20% relative	**+40% relative**
Consistency Rate	% output identity across repeated deterministic runs (`do_sample=False`)	100%	100%	**100%**
Hallucination Frequency	% of responses containing fabricated facts or SOP violations	<= 35%	<= 10%	**<= 2%**

Task 1.4: Design Methodology

`mermaid

flowchart TD

A[Customer Query] --> B{Intent Router LLM}

B -->|Solution V2: Fine-Tuned Qwen2.5| C[Extracted Intent JSON]

C --> D[ChromaDB Vector Retrieval]

E[13 Corporate SOP Documents] -->|Embedding: all-MiniLM-L6-v2| D

D -->|Top-k Relevant SOP Chunks| F[Context-Augmented Prompt Generator]

A --> F

F --> G[LLM Response Synthesizer]

G --> H[Final Policy-Grounded Customer Response]

`

1.4.1 Baseline Model Selection

• **Selected Architecture**: `Qwen/Qwen2.5-1.5B-Instruct` (Fallback: `TinyLlama/TinyLlama-1.1B-Chat-v1.0`).

• **Justification**:

• High performance on instruction-following and JSON formatting tasks.

• Parameter efficiency fits comfortably within single GPU T4 constraints (16GB VRAM) under 4-bit / 8-bit quantization.

• Native support for ChatML prompt formatting structures.

1.4.2 Retrieval Strategy Design

• **Embedding Model**: `sentence-transformers/all-MiniLM-L6-v2` (384 dimensions). Chosen for high retrieval accuracy and ultra-fast inference speed.

• **Vector Database**: `ChromaDB` persistent local vector store.

• **Chunking Strategy**: Markdown section header parsing and semantic paragraph split (chunk size ~300 tokens, overlap ~50 tokens).

• **Retrieval Workflow**:

1. Extract predicted `intent` from user query.

2. Perform cosine similarity vector search over SOP knowledge base using query + intent keyword enhancement.

3. Select Top $k=2$ highest relevance document chunks as context.

1.4.3 Fine-Tuning Strategy Design

• **Methodology**: Parameter-Efficient Fine-Tuning (PEFT) using **LoRA / QLoRA**.

• **LoRA Configuration**:

• Rank $r = 16$, $\alpha = 32$, Dropout = 0.05.

• Target Modules: `q_proj`, `k_proj`, `v_proj`, `o_proj`.

• **Training Setup**:

• Format: ChatML standard format (`<|im_start|>user ... <|im_end|><|im_start|>assistant ... <|im_end|>`).

• Loss Function: Cross-Entropy Loss with label masking (`-100` on prompt tokens) so loss is computed strictly on assistant output JSON tokens.

• Optimizer: `AdamW` (learning rate 2×10^{-4}, linear warm-up ratio 0.03, cosine decay).

• Validation & Early Stopping: Evaluation every 100 steps on active `val.jsonl` split; early stopping with patience 3 based on validation loss.

1.4.4 Evaluation Framework

• **Evaluation Dataset**: 600 held-out test examples (`test.jsonl`) with zero prompt overlap against training and validation partitions.

• **Adversarial Subset**: 100 adversarial / edge-case queries derived via regex filter (testing noisy inputs, typos, ambivalence).

• **Metrics Computation**:

• `Format Adherence`: `json.loads()` validation.

• `Intent Accuracy`: Exact string match on predicted `"intent"`.

• `ROUGE & BLEU`: `rouge-score` library and NLTK `sentence_bleu` against ground truth SOP excerpts.

• `Consistency`: Identity check across 3 deterministic passes (`do_sample=False`, `temperature=None`).

• `Hallucination`: Automated keyword assertion + policy contradiction checks.